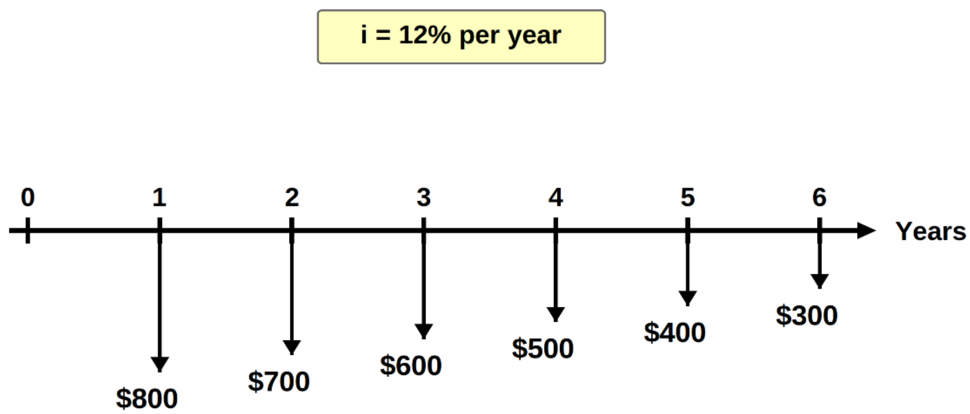
MUST	Misr University for Science & Technology	Final Exam – Spring 2025
	Faculty of Engineering Science & Technology	Level: 2nd Date: May 29th, 2025 Number of Questions: 6 No. of Pages: 2
	Department of Industrial Systems Engineering	**Exam Duration: two hours**
Total Marks: 60 Marks.	**Engineering Economics** **GE205**	**Exam Duration: two hours**

ملحوظة هامة: ابداء إجابة كل سؤال في صفحة جديدة من صفحات كراسة الإجابة

Question #1 (10 Marks)

Find the annual worth of the following cash flows at an interest of 12% per year.



Question #2 (10 Marks)

a) A company spent \$4,868 on a machine that provides an income of \$1,000 per year for 7 years. What is the rate of return on this investment?

b) If \$52,052 are deposited each year in a saving account that earns interest at a rate of 8% per year. What is the number of years required for the account to accumulate \$650,000?

Question #3 (10 Marks)

a) A milling machine was purchased for \$85,000. It is anticipated that its salvage value of \$15,000 will be realized after 10 years. During this time its average annual revenue totaled \$18,000 and annual operating and maintenance (O&M) cost was \$1,800 the first year and expected to increase by a constant amount of \$500 per year. Calculate its capital recovery cost at an interest rate of 10%.

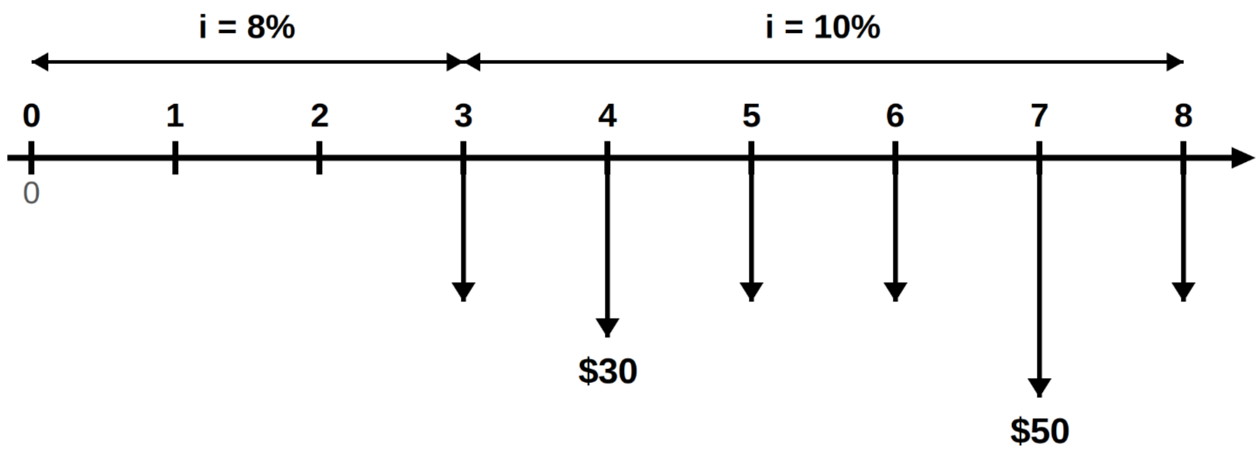
b) Determine the capitalized cost of \$10,000 every 5 years forever, starting 5 years from now at an interest rate of 12% per year.

Question #4 (10 Marks)

For the following cash flows, determine:

a) The present worth

b) The future worth



Question #5 (10 Marks)

Determine the present worth of the following cash flows at an interest rate of 16% per year compounded semiannually.

EOY	0	1	2	3	4	5
$	0	0	21,000	24,000	0	10,000

Question #6 (10 Marks)

A company owns a 5-year-old machine that was purchased for $75,000. The present market value of this machine is $35,000. Its annual operating cost is $12,000, and it will have $20,000 salvage after 3 years. A new equivalent machine can be purchased for $45,000. Its annual O&M cost is $9,000, and it will have $25,000 market value after 3 years.

a) Draw the cash flow for each alternative using the insider's view-point approach.

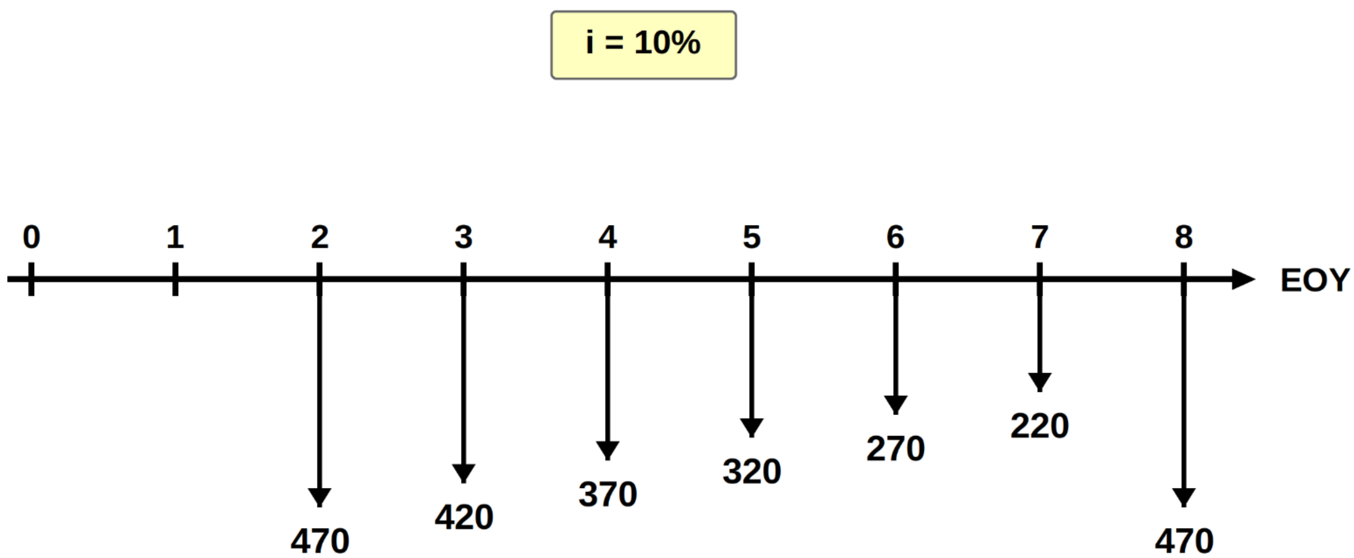
b) Using the present worth method and outsider's view-point approach, perform the replacement study at an interest rate of 10% per year.

Good Luck!

<table>
<tr><td rowspan="3">MUST</td><td>**Misr University for Science & Technology**</td><td>**Final Exam – Fall 2025**</td></tr>
<tr><td>**Faculty of Engineering Science & Technology**</td><td>Level: 2nd
Date: January 14th, 2026
Number of Questions: 6
No. of Pages: 8</td></tr>
<tr><td>**Department of Industrial Systems Engineering**</td><td>**Exam Duration: 2 hours**</td></tr>
<tr><td>**Total Marks: 60 Marks.**</td><td>**Engineering Economics**
GE205</td><td>**Exam Duration: 2 hours**</td></tr>
</table>

Question #1 (10 Marks)

Find the equivalent annual worth in years 1 through 8 for the following cash flow:



Annual Worth = ??

Question #2 (10 Marks)

Two machines that have the following costs are under consideration. Using an interest rate of 10% per year, determine which alternative should be selected on the based on present worth analysis.

	Machine A	**Machine B**
First cost, $	15,000	18,000
Annual O&M costs, $	3,500	3,100
Salvage value, $	1,000	2,000
Life, years	4	8

Question #3 (10 Marks)

a) (5 Marks) Determine the capitalized cost of constructing a dam at a cost of \$2,000,000, an annual operating cost of \$5,000, and periodic maintenance cost of \$10,000 every 5 years forever, starting 5 years from now at an interest rate of 12%.

b) (5 Marks) A milling machine was purchased for \$285,000. It is anticipated that its salvage value of \$50,000 will realized after 10 years. During this time its average annual revenue totaled \$52,000 and annual operating and maintenance (O&M) cost was \$10,000 the first year and increased by a constant amount of \$1000 per year. Calculate its capital recovery cost at an interest rate of 8%.

Question #5 (10 Marks)

For the following Cash flow diagram, it is required to find:

a) the present worth
b) the future worth
c) the equivalent interest rate over the 8-years period